

L^AT_EX-Workshop

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Cover

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Introduction

- What is $\text{\LaTeX}$?

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Introduction

- What is $\text{\LaTeX}$?
- What is it used for?
- What is it NOT?
- Why do we need to use it?

Syntax

- Designed to focus on writing
- Text will be output as text
 - White space is ignored (including newlines)
 - Empty line signifies end of paragraph
 - $\LaTeX$ automagically does hyphenation and outlining
 - “Magical” characters:
\$ \% ^ & _ { } ~ \backslash
 - Cannot use these in running text
- Comments start with a %

Syntax

- Commands start with a `\`:
`\somecommand[optional]{required}`
- Groups start with `{` and end with `}`
 - Useful if command influences all following text
This `{\bf}` text is bold} but this it not
This **text is bold** but this it not
- Environments start with `\begin{envname}` and end with `\end{envname}`
 - Define new commands that only work in environment
 - Change flow of text
 - Introduce special elements (i.e. non-text) in document

Editors

- There exist several $\text{\LaTeX}$ editors
- We generally recommend using one of the following editors below
- Online editor:
 - Overleaf (www.overleaf.com) **recommended**
- Offline editors (all cross platform):
 - Texmaker (www.xmlmath.net/texmaker)
 - TeXstudio (www.texstudio.org)
 - TeXworks (www.tug.org/texworks)

Overleaf

- If you're not familiar with $\text{\LaTeX}$ yet, we recommend you to use Overleaf during this tutorial
- You can register an account or login with your Google account at `www.overleaf.com`

Document Structure

- Document consists of two parts:
 - Preamble
 - Content
- Further forces $\text{\LaTeX}$'s focus on content

The preamble

- Set up for the document
 - Documentclass
 - Loading packages
 - Document wide settings
 - New commands/environments
 - Meta information (title, author, etc. . .)
- Allows us to change anything about the document
 - Also paper size, margins, etc
 - After preamble no longer possible

The document class

- Every document has document class
- Specifies general settings like paper size, margins, headers, footers, etc. . .
- Is set by `\documentclass` command:
`\documentclass[a4paper, 10pt]{article}`
- Different types of document classes:
 - article
 - beamer
 - report
 - book
 - letter
 - cover-article
 - . . .

Importing packages

- $\text{\LaTeX}$ files with commands/environments
- Can also be used to set setting (i.e. geometry)
- Extends $\text{\LaTeX}$'s usual capabilities.
- Imported with `\usepackage`
`\usepackage[hidelinks]{hyperref}`
- Contains all kinds of stuff:
 - `hyperref` for making hyperlinks
 - `graphicx` for importing images
 - `amsmath` for even better math support
 - `colorx` for changing colors
 - `tikz` for drawing figures in $\text{\LaTeX}$
 - `coffee` for adding coffee stains
 - `enigma` for confusing the allied forces

Setting settings

- Settings depend on packages used and needs of author
- `\title{}`, `\author{}` and `\date{}` are quite common

The content

- Content goes into document environment:

```
\begin{document}
```

...

```
\end{document}
```

- Normally starts with a title: `\maketitle`
- For longer documents a table of contents:
`\tableofcontents`

Document structure

- Layout is done by $\text{\LaTeX}$, structure is important
- Different kinds of headings:
 - `\chapter` only in book and report
 - `\section`
 - `\subsection`
 - `\subsubsection`
 - `\paragraph` typeset in running text
 - `\subparagraph`
 - Also a starred version to disable numbering: `\section*{}`
- `\appendix` changes number to appendix numbering

Text flow

- $\text{\LaTeX}$ ignores white space
- Works by splitting text up into tokens and paragraphs
- Tokens are separated by whitespace or punctuation
- Empty line ends paragraph
- Sometimes you want to force a space or newline
- For space, the non-breaking space: `~`
- For newline: `\\`
 - Warning! This does *not* start a new paragraph

Simple formatting

- Changing text size:

- `\tiny`
- `\scriptsize`
- `\footnotesize`
- `\small`
- `\normalsize`
- `\large`
- `\Large`
- `\LARGE`
- `\huge`
- `\Huge`

- **Bold text:**

`\textbf{text}` `{\bf text}`

- *Italic text:*

`\textit{text}` `{\it text}`

- *Emphasised text:*

`\emph{text}`

- Underlined text:

`\underline{text}`

- Monotype text:

`\texttt{text}` `{\tt text}`

Lists

- Three kinds of list:
 - ① `itemize` bulleted list like the outer list
 - ② `enumerate` numbered list like this one
 - ③ `description` list of definitions
- All lists are environments
- New items are added with `\item`
- Lists can be nested up to five levels deep
- Numbering happens automatically

Referencing

- $\text{\LaTeX}$ numbers a lot of stuff
- Referring back to parts of document is easy
- Name object with `\label{name}`
- Get number with `\ref{name}`
- Get the page number of object with `\pageref{name}`
- (With hyperref) Get name, number, and link
`\autoref{name}`
- Can be used to refer to:
 - Sections, subsections, etc. . .
 - Items in an enumeration
 - Equations
 - Figures
 - Tables

What are floats?

- $\text{\LaTeX}$ tries to structure text for optimal flow
- Does not work if we include figures and tables
- Solution: Floats
- Float is a container that cannot be broken over pages
- $\text{\LaTeX}$ moves floats through document for optimal placement
- Works good for images, tables, pseudocode, and other blocks
- Allows us to enrich content
 - Captions
 - Subfigures
- Can also define our own floats

Controlling float placement

- $\text{\LaTeX}$ tends to move floats all over the place
- Two ways to control floats:
 - Placement specifiers
 - Float barriers
- For reports: pictures at end is fine, lots of empty space is not

Controlling float placement

- $\text{\LaTeX}$ tends to move floats all over the place
- Two ways to control floats:
 - Placement specifiers
 - Optional argument to the float
 - Does not always work “correctly”
 - Pictures still float
 - `h` I want my float here
 - `H` Force float here (requires `float` package)
 - `t` I want my float at the top of the page
 - `b` I want my float at the bottom of the page
 - `!` I really want this
 - `h!` tells $\text{\LaTeX}$ you really want the float here
 - Float barriers
- For reports: pictures at end is fine, lots of empty space is not

Controlling float placement

- L^AT_EX tends to move floats all over the place
- Two ways to control floats:
 - Placement specifiers
 - Float barriers
 - Stops floats from moving past a point
 - Useful to keep floats within section
 - Require the `placeins` package
 - Created with command `\FloatBarrier`
 - Automatic float barriers at sections:
`\usepackage[section]{placeins}`
- For reports: pictures at end is fine, lots of empty space is not

Working with images

```

\begin{figure}[htb]
  \includegraphics[width=\textwidth]{path/to/file}
  \caption{Something sensible about this image}
  \label{fig:my_image}
\end{figure}

```

- [htb] is placement specifier
- `\includegraphics` for importing of the image
- `\caption` for text below image and numbering
- `\label` so we can refer later

Working with tables

```

\begin{table}
  \caption{Caption above table}
  \label{tbl:square}
  \begin{tabular}{|c|c|c|}
    \hline
    2 & 7 & 6 \\
    \hline
    9 & 5 & 1 \\
    \hline
    4 & 3 & 8 \\
    \hline
  \end{tabular}
\end{table}

```

Table: Caption above table

2	7	6
9	5	1
4	3	8

Three useful sites:

- en.wikibooks.org/wiki/LaTeX/Tables
- www.latex-tables.com
- www.tablesgenerator.com

Working with floats

- $\text{\LaTeX}$ knows the best way to position floats
- Bunch of pictures at the end of report not bad
 - Graders know why
 - Large sections of white space are more annoying
- For figures: changing the width so two pictures fit on page
- Text on both sides of float help with positioning
 - More text $\Rightarrow$ more things for $\text{\LaTeX}$ to play with
- Float barriers can help if document is broken up in logical sections
 - E.G. all the figures for exercise 1 stay with exercise 1

Mathematics

- Most of the following commands need the package `amsmath`
- `$ ... $` or `\( ... \)` short hand for inline math
- `$$ ... $$` or `\[ ... \]` short hand for display math mode
- The `equation` environment displays numbered equations
- The `align` environment is used to arrange equations over multiple lines. `\\` denotes a new line and `&` indicates a point where the equations should be aligned
- Spaces are derived from the expressions, `$2-3$` and `$2 - 3$` both print to $2 - 3$

Common commands

- `\times` is $\times$, `\cdot` is $\cdot$
- `\sqrt{a}` is $\sqrt{a}$
- `\frac{a}{b}` is $\frac{a}{b}$
- `a^{b}` is a^b
- `a_{b}` is a_b
- All greek letters are available. `\Delta` is Δ and `\delta` is δ
- Text can be added to the equation with `\text{}`.
 $2 - 3 \text{ hallo } 3 \times 4$ is $2 - 3 \text{ hallo } 3 \times 4$

Operators

- `\cos (2\pi)` is $\cos(2\pi)$
- `\sin^2 (2\theta)` is $\sin^2(2\theta)$
- `\tan (2)` is $\tan(2)$
- `\lim_{x \to \infty} \exp(-x) = 0` is $\lim_{x \rightarrow \infty} \exp(-x) = 0$
- `x \equiv a \pmod{b}` is $x \equiv a \pmod{b}$
- <https://en.wikibooks.org/wiki/LaTeX/Mathematics> lists more commands
- detexify.kirelabs.org can identify a latex symbol from a drawing

Why chemists use TeX

- Chemistry writing needs consistent notation for formulas, reactions, units, and references
- $\text{\LaTeX}$ is strong at keeping that notation uniform across long reports and theses
- Two packages show up all the time in chemistry documents:
 - `mhchem` for chemical formulas and reactions
 - `siunitx` for numbers, units, and uncertainties
- The goal is readable source and publication-quality output

Chemical formulas and reactions

- `\ce` formats formulas, ions, and reactions
- Examples:
 - Water: H_2O
 - Oxidation state: Fe^{2+}
 - Reaction: $2 \text{H}_2 + \text{O}_2 \rightarrow 2 \text{H}_2\text{O}$
 - Equilibrium: $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$
- This keeps subscripts, charges, arrows, and stoichiometry consistent

Units, quantities, and data

- `siunitx` keeps units separated from values
- Examples:
 - Volume: `\SI{25}{\milli\liter}`: 25 mL
 - Temperature: `\SI{298.15}{\kelvin}`: 298.15 K
 - Concentration: `\SI{1.20e-3}{\mole\per\liter}`:
 $1.20 \times 10^{-3} \text{ mol L}^{-1}$
 - Count with formatting: `\num{6.022e23}`: 6.022×10^{23}
- Useful for lab reports, tables, and plots where significant figures matter

Common chemistry document tips

- Put reaction schemes, spectra, and tables in floats so they can be captioned and referenced
- Use `\label` and `\ref` for compounds, schemes, and figures
- Keep citations in a `.bib` file so experimental sources and papers stay manageable
- If your report has a lot of notation, define a few small commands in the preamble to reduce repetition

Source Code

- $\text{\LaTeX}$ can be used to format source code
 - Requires `listings` package
- Can include (parts of) document, snippets, and lines
- Automatic syntax recognition
- Allows different styles
 - Most languages have settings online
- `\lstinputlisting{file}` to include file
- `\begin{lstlisting}` and `\end{lstlisting}` for snippets
- `\lstinline{code}` for inline code

Settings

listings offers several settings for each command, can be set globally using `\lstset{}`

Setting	Values	Explanation
breaklines	true/false	If code should be broken over lines
breakatwhitespace	true/false	If code can only break at whitespace
language	string	The language the code is written in
numbers	none/left/right	Where to position the line numbers
stepnumber	number	The numbers in between two steps

Bibtex

Useful package for referencing sources.

- Stores references in separate ".bib" file.
 - Set style with `\bibliographystyle{plain}`
 - Include bib file with `\bibliography{filename}`
 - These should be placed in this order at the location you want the reference list to be printed
- `\cite{}` to cite the references in the bib file
- More information at https://en.wikibooks.org/wiki/LaTeX/Bibliography_Management.

General

Package	Description
<code>enumitem</code>	Changes the symbols used in enumerations
<code>babel</code>	For writing text in languages that are not English
<code>fancyhdr</code>	For making headers and footers
<code>tikz</code>	For the creation of drawings in $\text{\LaTeX}$
<code>moderncv</code>	For making a résumé

Linguistics

Package	Description
<code>tipa</code>	For symbols from the IPA
<code>qtree</code>	For drawing (syntactic) trees
<code>tikz-qtree</code>	For drawing (syntactic) trees with <code>tikz</code>
<code>gb4e</code>	For glosses and the likes
<code>phonrule</code>	For phonological rules
<code>lplfitch</code>	For typesetting natural deduction proofs